Introduction to University Mathematics

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Abstract

Notes for the University of Helsinki course MAT11001, taught in Finnish from 31 August to 17 October 2026.

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1 Course information

The official course information is available on the University of Helsinki course page.

2 Lecture notes

2.1 31 August 2026

Hello, World!

Definition 2.1. An integer $p > 1$ is a *prime number* if its only positive divisors are 1 and p.

Lemma 2.2. *Every integer greater than* 1 *has a prime divisor.*

Proof. We use strong induction. Let $n > 1$, and suppose the claim holds for every integer strictly between 1 and n. If n is prime, then it divides itself. Otherwise, $n = ab$ for some integers $1 < a < n$ and $1 < b < n$. By the induction hypothesis, a has a prime divisor, which then also divides n. $\square$

Theorem 2.3. *There are infinitely many prime numbers.*

Proof. Suppose instead that $p_1, \ldots, p_k$ are all the prime numbers, and set

$$N = p_1 p_2 \cdots p_k + 1.$$

By lemma 2.2, some prime q divides N. Our assumption gives $q = p_i$ for some i, so q divides both $p_1 \cdots p_k$ and N. It must therefore divide their difference 1, which is impossible. $\square$

Corollary 2.4. *For every positive integer n, there is a prime number greater than n.*

Proof. Otherwise, all prime numbers would belong to the finite set $\{2, \ldots, n\}$, contradicting theorem 2.3. $\square$